

Assessing Learned Models for Phase-only Hologram Compression

Zicong Peng
KOC University
Istanbul, tr
zpeng25@ku.edu.tr

Kaan Akşit
University College London
London, UK
k.aksit@ucl.ac.uk

Abstract

Based on what I have learnt so far, I would like to systematically organize the papers, researches, classes, books that I found and explored. This paper will try to collect the recent Models, perspectives and theories in good conference or journals. In addition, it is the note to record my idea and question to improve my understanding and probably prove my directions

CCS Concepts

• **Computing methodologies** → **Image compression**; • **Theory of computation** → **Data compression**; • **Hardware** → *Displays and imagers; Emerging optical and photonic technologies.*

Keywords

eyes tracking, deep learning, computer graphics

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1 Introduction

example introduction

2 Theory

2.1 eyes tracking

2.1.1 background. To find the beginning of this topic, I always wanna how to describe what we can see in real world and also what kinds of factors would influence them, so I find that Watson and Yelot (2012) conducted a systematic and comprehensive study to explore the effects of visual conditions and other factors (monocular vision and binocular vision, brightness, field size, age, adaptation status, cognitive load, etc.) on the pupil's light response. To be more specific, this is waht they explore

$$D_{sd}(F) = 7.75 - 5.75 \times \frac{(F/846)^{0.41}}{(F/846)^{0.41} + 2} \quad (1)$$

where:

- $D_{sd}(F)$ represents the purely physiological target pupil diameter, the size of which is driven entirely by the physical properties of ambient light.

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- F denotes the effective retinal illuminance, measured in Trolands (Td).

The Watson-Yellott pupil dynamics formula (Equation ??) functions as the system's internal "light meter." Within the `lens_deform.comp` module, this equation is utilized to convert the intensity of the external stimulus light source into effective retinal illuminance, thereby calculating the target physiological pupil diameter strictly as a function of environmental lighting conditions. Based on that, to the next step, [?] demonstrate that binocularity and eye-dominance-modulated retinal spatial non-uniformities jointly shape the natural nonlinear variations in visual-motor tracking, providing a crucial baseline for understanding visual neural mechanisms and clinically detecting neuro-ophthalmological impairments.

2.1.2 simulation. Next step, I also find some equation or method to construct pupil expense and reflection in lens

- **Asymmetric Damped Spring Filter:**

$$\alpha = 1 - \exp\left(-\frac{dt}{\tau}\right)$$

It simulates muscle dynamics, giving the pupil's light response realistic latency and biological inertia.

- **Paraxial Mirror Equation:**

$$v = \frac{uR}{2u - R}$$

It bypasses computationally expensive ray tracing to directly calculate the precise 3D coordinates of the Purkinje reflections.

- **Schlick's Fresnel Approximation:**

$$F(\theta) = R_0 + (1 - R_0)(1 - \cos \theta)^5$$

It rapidly computes surface reflectance based on viewing angles, perfectly recreating the bright corneal and dim lenticular optical layers.

3 Good model paper

This article [?] emphasizes the need to bring the model back to the first principles, is, "let the denoising model truly perform denoising". Firstly, based on the "manifold assumption", it is fundamentally inconsistent with the characteristics of noise (ϵ) and flow velocity (v). However, in fact, predicting a clean image only requires the network to retain low-dimensional information and filter out the noise. I think the advantage of this article lies in pointing out that the current diffusion models based on pixel space (Pixel-space) and the ViT architecture are overly obsessed with calculating high-dimensional information. I also believe that these methods are overly complex and difficult to understand and further optimize except for continuously increasing computing power. Therefore, this article's proposal of a sufficiently simple idea is worth reading. The problem is also

obvious The article mentions that the quality of generated images is not as good as the currently developed pre-trained models. Since he mentioned this, I think it should more than what he said. Moreover, the model structure is too simple, and I think the old problem of overfitting should also occur [?]. In addition to simplify high dimensional data, this paper also would like to explore form latent space optimization in terms of srable diffusion. In summary, This article aims to use a powerful pre-trained visual representation encoder to replace the traditional VAE. To be more specific, The main DiT heavily relies on the traditional VAE to compress. However, high compression contian potential loss in low dimation, to keep these information this paper propose frezzing pretrain model DINOv2, which use additional vit to transfer token to pixel Moreover, DiTDH is a wide and shallow network. Therefore, the performance could beyond SD-VAE. So in my opinion, it is also a good way to ispire us use a new direction in find the better bottleneck of vae straccture and improve model's information retrieval capability in the good way, this method reduce the difficulty of compressing and achieved a obvious better convergence speed On the other hand, it also need add noise to improve the decoder's reconstruction in short, it loss precise reconstruction ability to some extent [?].

my question is Will this method undermine the already powerful retrieval ability of the transformer?

4 Introduction

4.1 plan

The precise prediction of accommodation and vergence heavily depends on the quality of the synthesized ocular reflections. Historically, researchers have employed Zemax for such optical simulations; however, we found that traditional ray-tracing software is computationally exhaustive and yields suboptimal visual fidelity for dynamic Purkinje image generation.

To optimize both rendering time and optical effect, we constructed our simulation pipeline using Blender's advanced rendering engine. By transitioning from conventional PCCR—which infers accommodation from fixation bias—to a DPI-centric approach within this modern rendering framework, we achieved superior optical sampling rates. The updated software allows for more efficient multi-path light tracing, enabling the extraction of high-resolution Purkinje images that accurately represent the eye's physical accommodation state.

4.2 In-vivo Data Acquisition and Multimodal Inference

Building upon our established hardware framework, we extend our methodology to capture empirical in-vivo data from human subjects. To bridge this physical hardware with state-of-the-art computational paradigms, we conceptualize an intelligent diagnostic agent capable of automated clinical evaluation and the generation of plausible synthetic datasets for both medical and consumer applications. As a proof-of-concept for this integration, we formulate a novel inferential pipeline: raw optical captures acquired directly from the hardware are fed into a Multimodal Large Language Model (MLLM). Rather than relying solely on traditional analytical algorithms, we leverage the MLLM's emergent reasoning capabilities to infer critical ocular properties, estimating physiological metrics

such as vergence and dynamic accommodation states directly from the visual inputs.

5 Results and Discussions

example result

example table

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